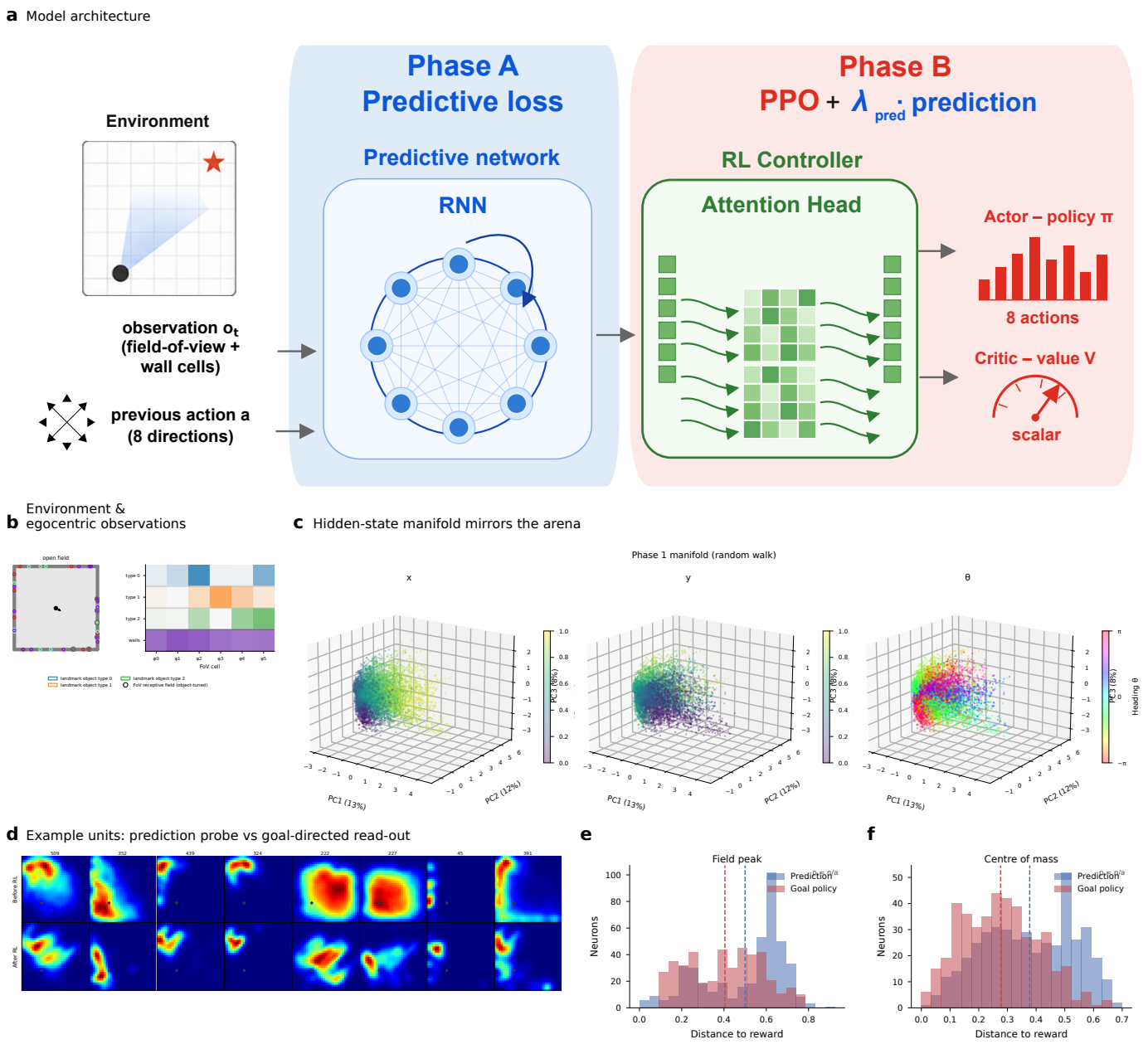

a Model architecture
Environment
observation o_t (field-of-view + wall cells)
previous action a (8 directions)
Phase A
Predictive loss
Predictive network
RNN
Phase B
PPO + λ_{pred} · prediction
RL Controller
Attention Head
Actor – policy π
8 actions
Critic – value V
scalar
b Environment & egocentric observations
open field
FoV cell
landmark object type 0
landmark object type 1
landmark object type 2
FoV receptive field (object-tuned)
c Hidden-state manifold mirrors the arena
Phase 1 manifold (random walk)
x
y
θ
PC1 (13%)
PC2 (12%)
PC3 (8%)
Heading θ
d Example units: prediction probe vs goal-directed read-out
Before RL
After RL
e
Field peak
Prediction
Goal policy
Neurons
Distance to reward
f
Centre of mass
Prediction
Goal policy
Neurons
Distance to reward